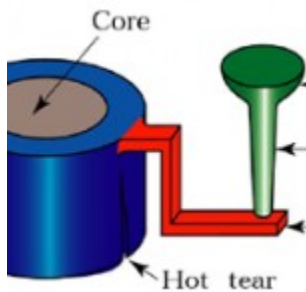


# Mechanical Engineering

## Defects In Casting | Molding Defects | Casting Defects And Remedies



### **Sand Casting Defects:**

- Production of castings involves a large number of steps including casting design, pattern making, moulding, melting, pouring, shake out, fettling, inspection and finishing.
- It is not uncommon for one or more of these steps to be performed unsatisfactorily due to use of defective material or equipment, carelessness of the operator or lack of skill.
- Such unsatisfactory operations result in a defective casting which may be rejected at the final stage.
- Since reclamation of defective castings is often costly and sometimes outright impossible, care should be taken to avoid the occurrence of the defects in the first instance.
- It is therefore necessary to understand the various defects that occur in sand castings and the main factors that are responsible for their occurrence.
- Some of the common defects are described below.

1. Open Blows and Blow Holes

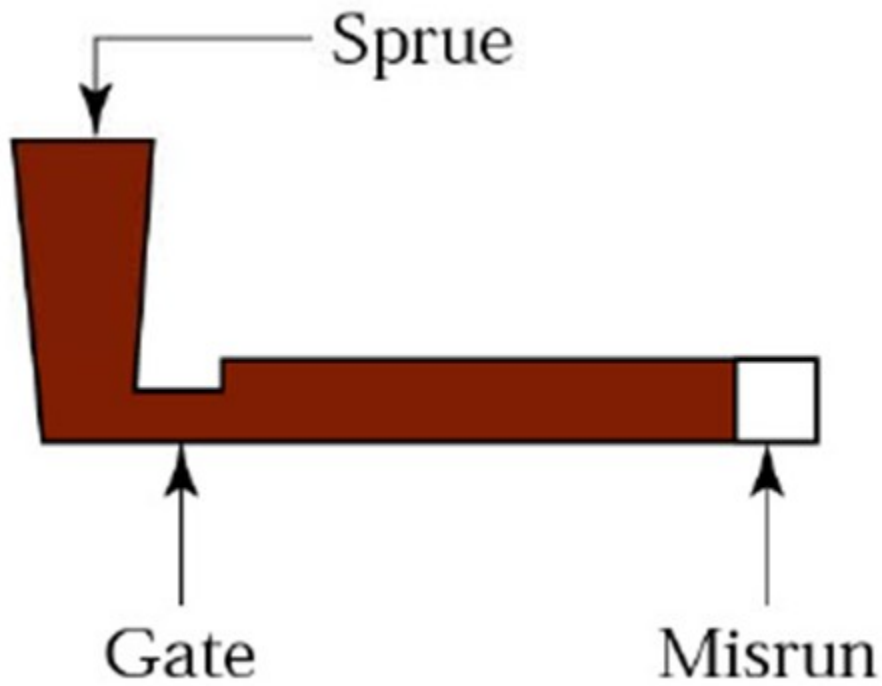
2. Pin Hole Porosity

3. Entrapped Air and other gases

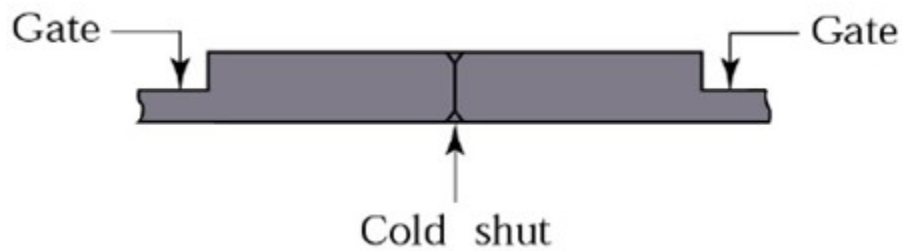
4. Cracked Casting
5. Bent or Twisted Casting
6. Dropped Mould
7. Fusion
8. Swell
9. Run out
10. Mismatch
11. Mis-run and Cold Shut
12. Shrinkage-Faults
13. Rat Tail and Buckles
14. Core Shift
15. Inclusions
16. Cuts and Washes
17. Metal penetration
18. Hard Spots
19. Scabs
20. Hot tears

**Misrun and Cold Shut:**

- A misrun is caused when the section thickness of a casting is so small or the pouring temperature so low that the entire section is not filled before the metal solidifies.



- Cold shut is caused when two streams of metal which are too cold meet but do not fuse together.



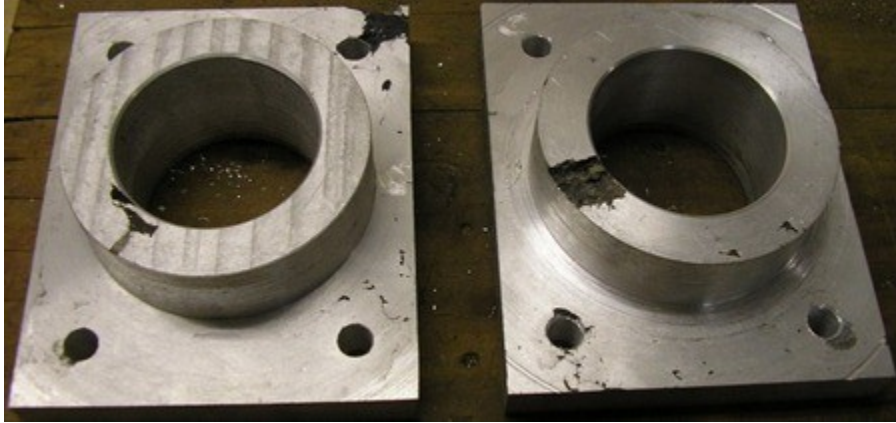
- Misrun and cold shut can be minimized by proper design of casting, providing suitable gating and risering and using correct temperature of the melt.



**MIS RUN**

**Shrinkage Faults:**

- Shrinkage faults are faults caused by improper directional solidifications, poor gating and risering design and inadequate feeding.
- Solidification leads to volumetric contraction which must be compensated by feeding. If this compensation is inadequate either surface shrinkage or internal shrinkage defects are produced making the casting weaker.
- Shrinkage faults can be reduced by providing proper gating system, pouring at correct temperature and taking care of directional solidification.



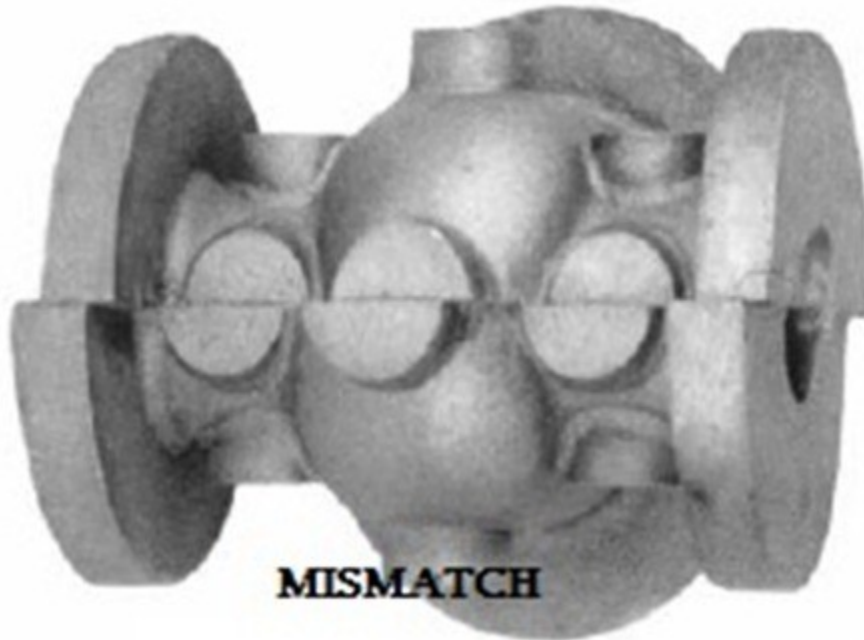
### **Rat Tail and Buckles:**

- Rat tails and buckles are caused by the expansion of a thin outer layer of moulding sand on the surface of the mould cavity due to metal heat.
- A rat tail is caused by depression of a part of the mould under compression which appears as an irregular line on the surface of the casting.
- A buckle is a more severe failure of the sand surface under compression.
- The mould must provide for proper expansion instead of forming compressed layers to avoid this defect.



**Core Shift:**

- A core shift results from improper support or location of a core.
- It results in a faulty cavity or hole in the casting.
- It can be reduced by providing proper support for cores and correct alignment with the mould.

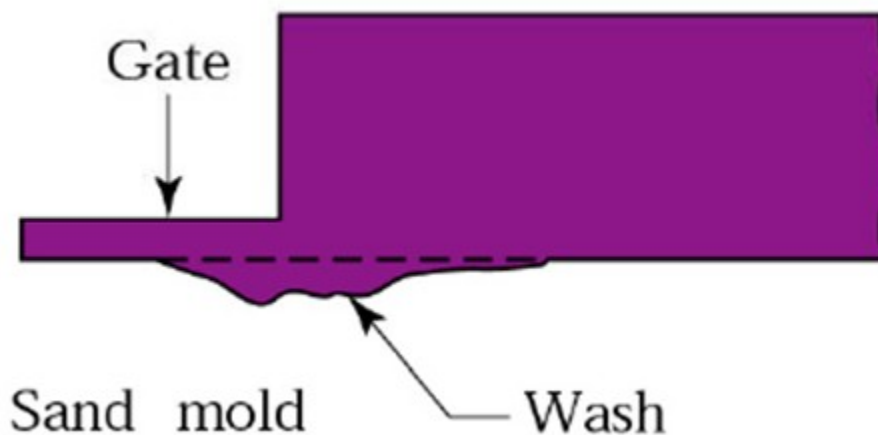
**Inclusions:**

- Inclusions are any foreign materials present in the cast metal.
- These may be in the form of oxides, slag, dirt, sand or nails.
- Common sources of these inclusions are impurities with the molten metal, sand and dirt from the mould not properly cleaned, break away sand from mould, core or gating system, gas from the metal and foreign items picked on the mould cavity while handling.
- Inclusions are reduced by using correct grade of moulding sand and proper skimming to remove impurities.



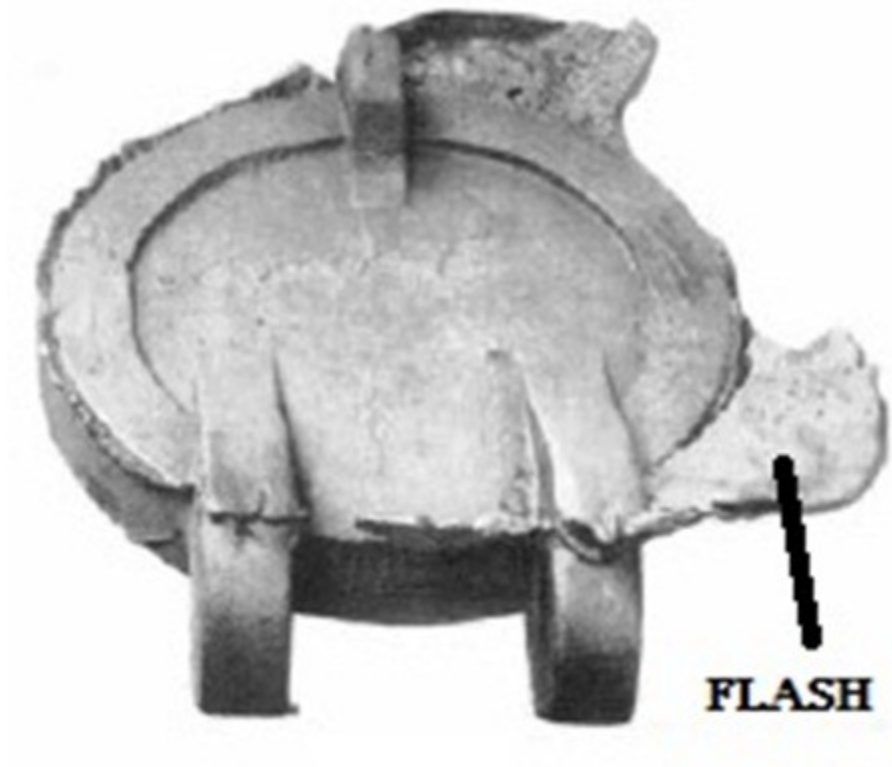
#### **Cuts and Washes:**

- Cuts and washes are caused by erosion of mould and core surfaces by the metal flowing in the mould cavity.
- These defects are avoided by proper ramming, having sand of required strength and controlling the turbulence during pouring.



#### **Metal penetration:**

- If the sand grains used are very coarse or the metal poured has very high temperature the metal is able to enter the spaces between sand grains to some distance. Such sand becomes tightly wedged in the metal and is difficult to remove.
- The remedy is to remove the causes mentioned above.

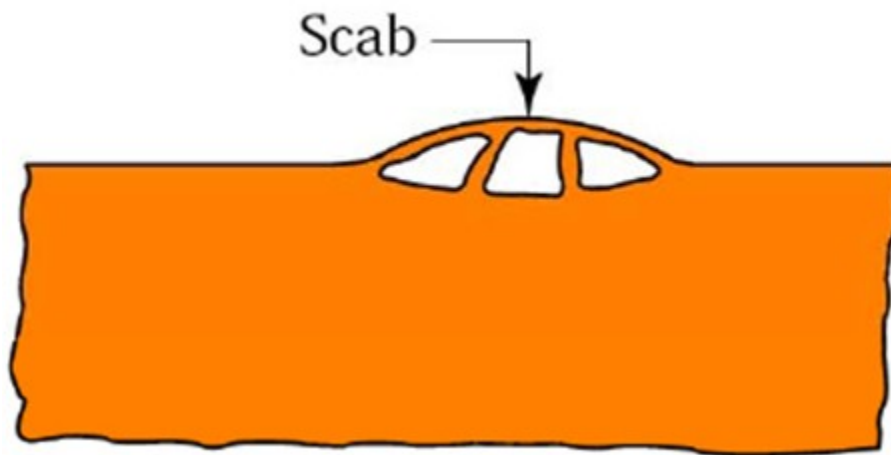
**Hard Spots:**

- Hard spots are caused by chilling action of moulding sands in some metals like gray cast iron with insufficient silicon.
- These spots are extremely hard and often lead to machining difficulties.
- Hard spots are avoided by providing uniform cooling and pouring at the right temperature.

**Scabs:**

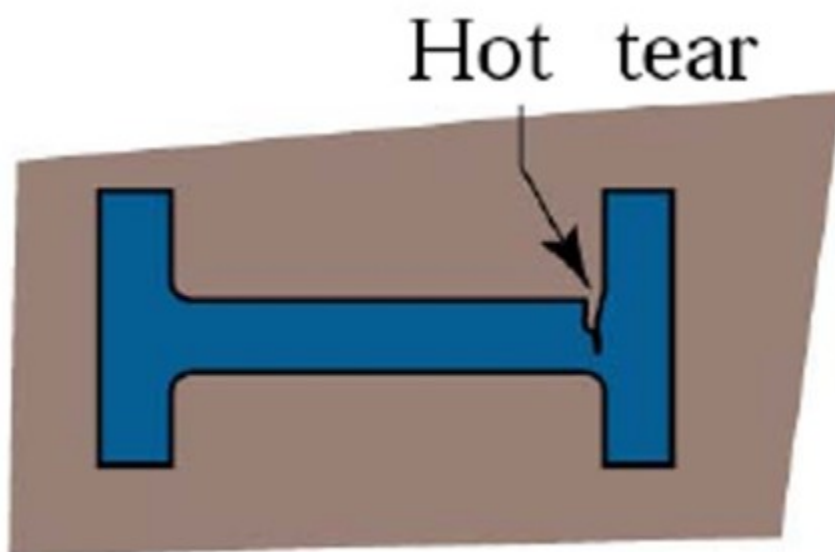
- Scabs are rough, irregular projections on surface of castings containing embedded sand.
- Scabs occur when a portion on the face of mould or core lifts and metal flows underneath in a thin layer.
- They are caused by using too fine sand grains or using sand of low permeability or moisture content.
- They may also be caused by uneven mould ramming or by intense local overheating.

- Scabs can be reduced by mixing additives like sea coal, wood flour or dextrin in the sand, providing uniform ramming and pouring with correct velocity.

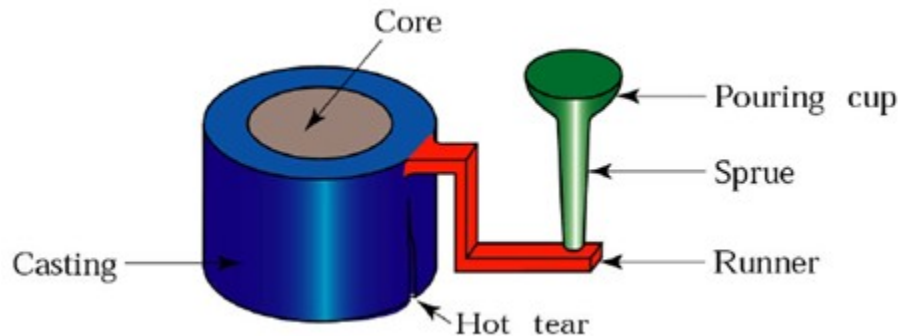


#### Hot tears:

- Hot tears are ragged irregular internal or external cracks occurring immediately after the metal have solidified.
- Hot tears occur on poorly designed castings having abrupt section changes or having no proper fillets or corner radii. Wrongly placed chills.
- Improper placement of gates and risers or incorrect pouring temperatures can also produce hot tears.



- Hot tears are also caused by poor collapsibility of cores.
- If the core does not collapse when the casting is contracting over it stresses will be set up in the casting leading to its failure.
- Hot tears can be eliminated by improved design, proper directional solidification, and uniform rate of cooling, correct pouring temperature and control of mould hardness.

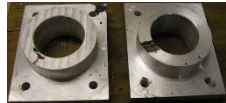


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## [Sand Casting Defects | Aluminium Casting Defects | Steel Casting Defects](#)



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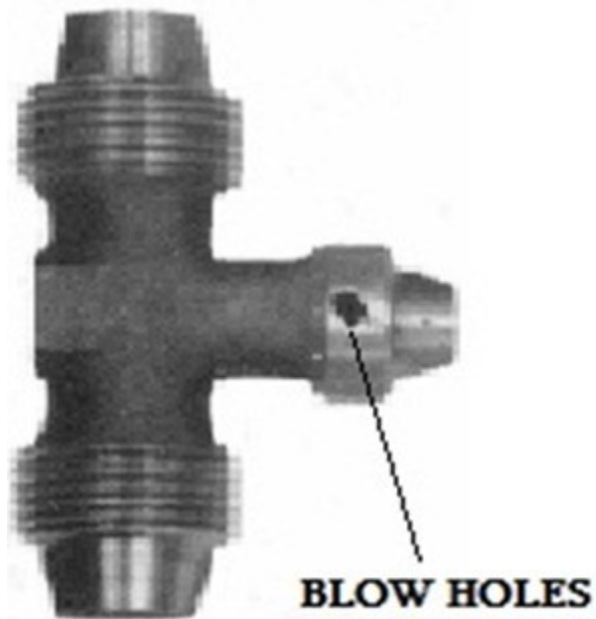
#### 1. Open Blows and Blow Holes

#### 2. Pin Hole Porosity

3. Entrapped Air and other gases
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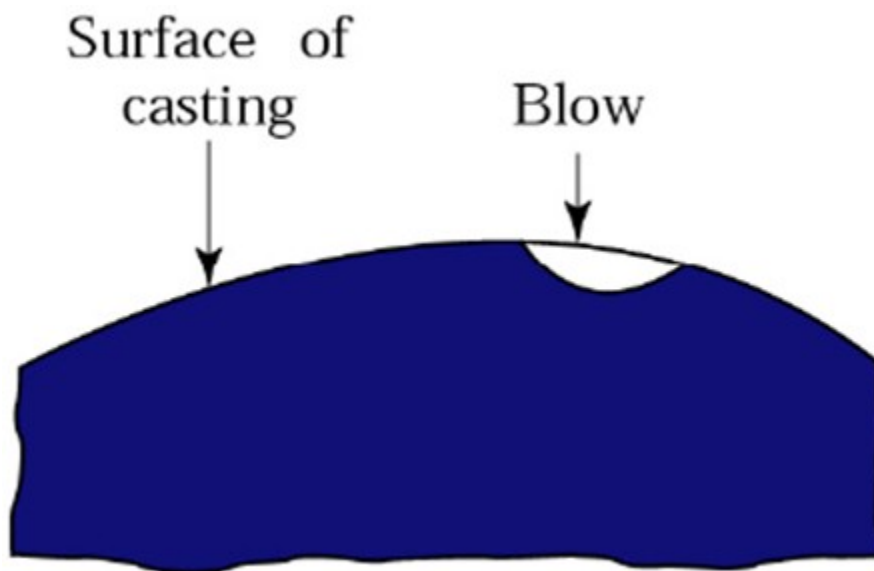
**Open Blows and Blow Holes:**

- Open blows are smooth cavities or voids on the surface of the casting. Blow holes are bubbles of gas entrapped inside the casting. Both are caused by gases carried by hot metal.



- Blow holes are also caused by poor permeability of sand, too fine sand grains, over rammed moulds or insufficient venting.

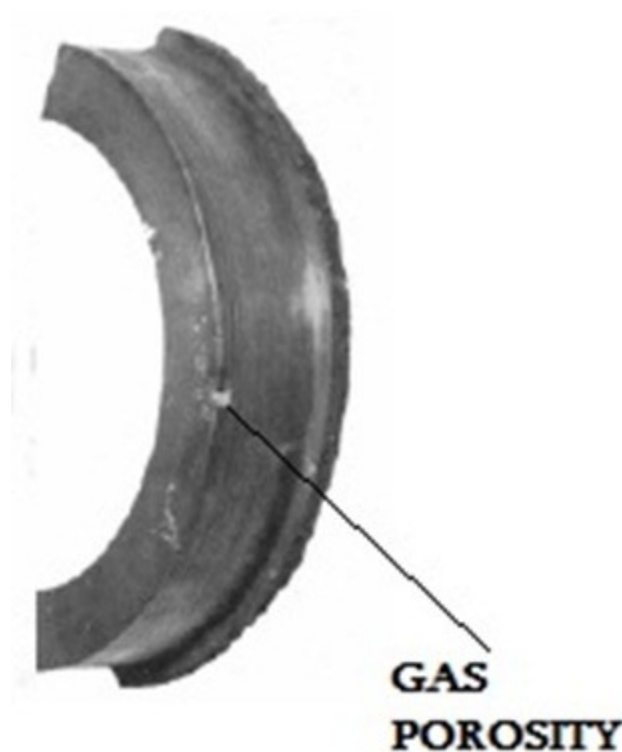
Blow holes can be controlled by controlling moisture content, providing proper venting, not too hard ramming of the mould and controlling binder content.



#### Pin Hole Porosity:

- This is porosity caused by gases absorbed by the molten metal.
- Practically all metals absorb oxygen, hydrogen and nitrogen.

- Oxygen and nitrogen form oxides and nitrides respectively. Hydrogen is responsible for causing pin hole porosity.
- Molten metal picks up hydrogen from fuel, moisture in air and moulds.
- As the metal solidifies solubility of hydrogen decreases considerably. The hydrogen thus comes out forming a number of small holes distributed throughout the metal.



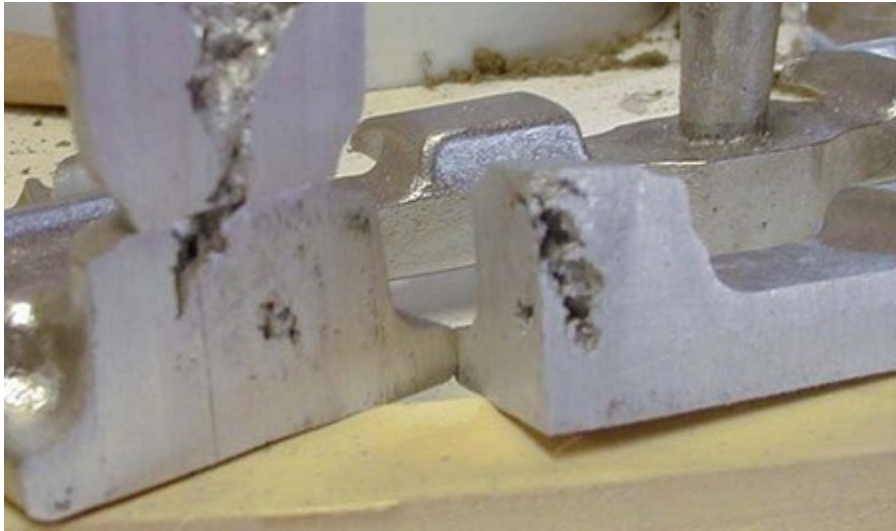
- Pin hole porosity is quite common in aluminium alloys.
- Pin hole porosity can be minimized by using improved melting and pouring methods are by prompting a rapid rate of solidification.

#### **Entrapped Air and other gases:**

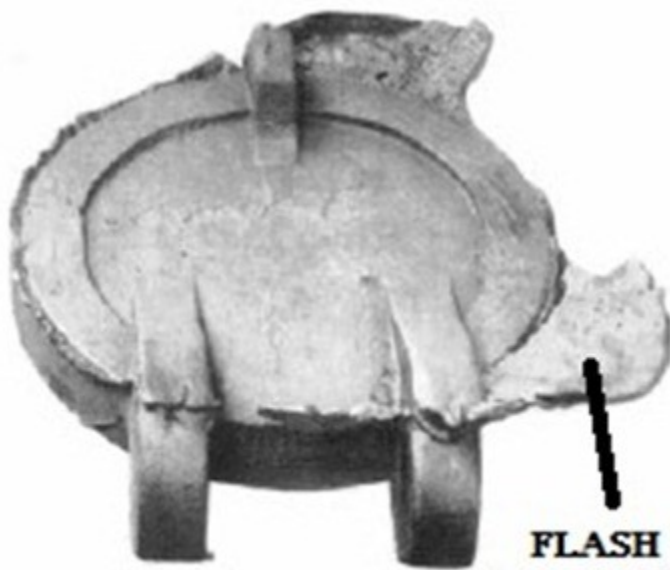
- Air or other mould gases are sometimes entrapped within the metal due to turbulent flow of metal during pouring.
- When the metal solidifies this air or gas remains in the metal as a void.
- This defect can be eliminated by providing proper gating systems and avoiding turbulence.

### Cracked Casting:

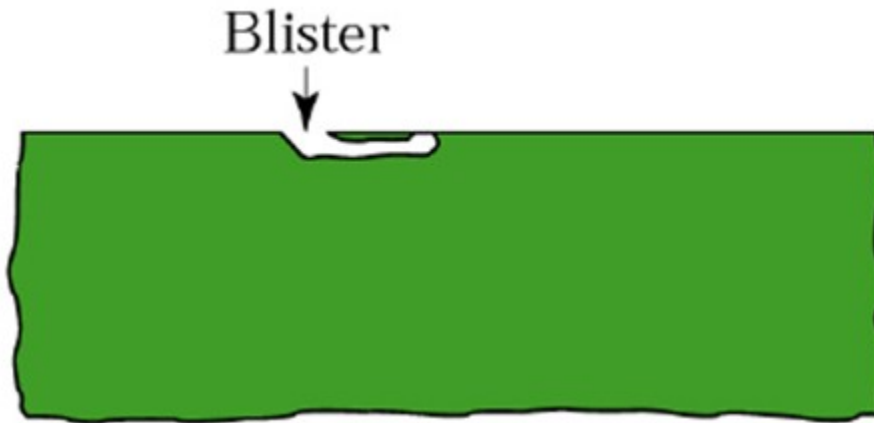
- These are cracks caused by the inability of the casting to cope up with internal stresses resulting from solid shrinkage of the metal.



- Immediately after solidification some metals have a low strength and if stresses set up are high, cracks develop in the casting.



- Cracks can be avoided by proper design of the casting, avoiding joining of unequal sections, providing round corners etc.



#### **Bent or Twisted Casting:**

- When a casting has wide variation in its sections, internal stresses are often set up in it due to differential cooling in the thick and thin areas. This leads to casting warpage.
- The remedy is to provide suitable allowances on the pattern, controlling deformation by providing suitable ribs etc. and plan for uniform cooling rate.



#### **Dropped Mould:**

- This defect occurs due to a portion of the mould breaking and dropping into the metal.

- This may occur when the sand mix has a low green strength or it has not been properly rammed or reinforced.
- It appears as an irregular deformation of the casting.
- It can be avoided by proper ramming and reinforcements of the mould.

#### **Fusion:**

- When the mould sand does not have enough refractoriness or the metal is poured at very high temperature or the facing sand is of poor quality, the sand may melt and fuse with casting surface.



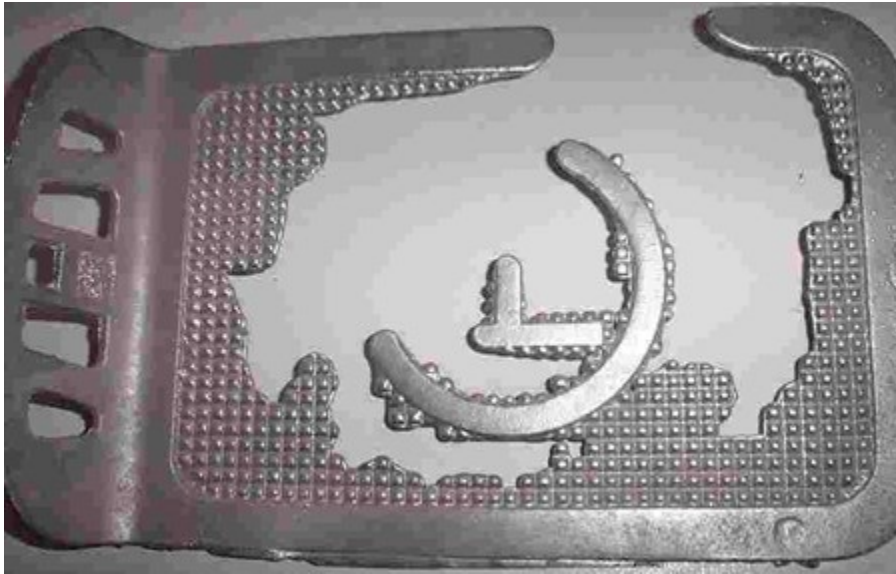
- This makes it difficult to clean the castings and gives them a rough glossy appearance.
- Using sand of enough refractoriness and pouring at suitable temperature can eliminate this defect.

#### **Swell:**

- A swell is an enlargement or bulging of the casting surface resulting from liquid metal pressure.
- It occurs due to poor ramming of the mould or not properly reinforcing deep moulds.
- Swells can be avoided by proper ramming of the sand and providing adequate support to the mould.

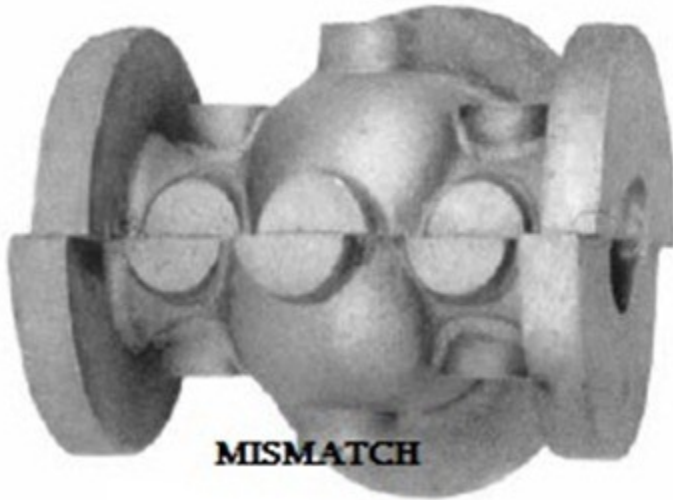
#### **Run out:**

- A run out occurs when the molten metal leaks out of the mould during pouring. This is caused by defective moulding boxes, faulty clamping and weighting and by casting too near the edge of the box.
- Run out is avoided by modifying the moulding system, providing proper parting line and providing proper gating system.



**Mismatch:**

- A mismatch is caused by the cope and drag parts of the mould not remaining in their proper position.
- This is caused by loose box pins, inaccurate pattern dowel pins or carelessness in placing the cope on the drag.
- The deficiencies mentioned above must be attended to if mismatch is to be avoided.

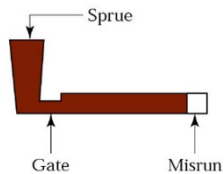


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